Machine Learning Project Report

To Grant or Not to Grant - Deciding on Worker's Compensation Benefits

Group 53
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Nova IMS | Fall/Spring Semester | December 2024

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Abstract

The New York Workers' Compensation Board (WCB) processes and evaluates claims related to workplace injuries, disability benefits, and volunteer workers' compensation. Since 2000, the WCB has reviewed over 5 million claims, demonstrating the significant workload and complexity of its processes. In this project, we aimed to develop a machine learning model to automate the decision-making process, specifically predicting the ***Claim Injury Type*** variable from a robust dataset spanning from 2020 to 2022. By leveraging advanced classification techniques, we explored multiclass prediction and optimization strategies, ensuring the model's scalability and generalizability. The results demonstrated significant potential to streamline WCB's operations, with the promise of reducing manual intervention and improving decision accuracy. Furthermore, the analysis provided additional insights into feature importance, highlighting critical factors that influence claims processing.

1. Introduction

This project is a classification problem with the goal of predicting the **Claim Injury Type** for workers' compensation claims, using data from 2020 to 2022, to automate decision-making and improve WCB processes. In this project, we took an in-depth analysis of the dataset with a focus on key attributes such as injury details, claimant demographics, and workplace information, which provided valuable insights into the patterns and relationships within the data. First, we tried to carefully clean and preprocess to prepare the whole dataset so that in our modelling and exploratory data analysis part, it can be performed better. Then we built and applied several machine learning models using advanced classification techniques, fine-tuning and optimizing them to maximize their performance. In this process, we developed a model capable of accurately predicting injury types, reducing manual intervention in the claims review process. In this report, we will discuss the steps and the results of our analysis that we took to complete this project.

2. Exploration

During the data exploration phase, we tried to gain a deeper understanding of the patterns and relationships within the claims data submitted to the New York Workers' Compensation Board (WCB). To do this, we undertook the following steps:

2.1. Examining the Dataset

The dataset contains 593,471 rows and 33 columns with a mix of numerical and categorical variables. Using head() and info(), we identified 11 numerical variables, such as Age at Injury and IME-4 Count, stored as float64 or int64, representing continuous or discrete values. Additionally, there were 22 categorical variables stored as objects, indicating they hold string-based or category-like data, including features such as District Name, Gender, and Injury Descriptions.

2.1.1. Comprehensive Data Cleaning: Missing Values, Duplicates and Data Imputation

During the analysis of missing values and duplicates, we identified several columns with significant missingness and addressed them systematically. Using the isnull() and isna() methods, we detected nullable and missing values in key columns such as IME-4 Count, First Hearing Date, and C-3 Date. Notably, we found these columns exhibited over 60% missing values, but instead of removing them outright, we retained them as their missingness showed strong correlations with the target variable, Claim Injury Type. We applied binary encoding to capture the presence or absence of values in these columns, which revealed important patterns. For instance, missing values in IME-4 Count and First Hearing Date were associated with simpler claims like NON-COMP, while the presence of values corresponded to more complex claims such as TEMPORARY or PPD SCH LOSS. Additionally, hidden missing values in non-metric features, such as Carrier Type and Gender, we standardized them by converting placeholder values like UNKNOWN and U to NaN. We removed rows with ambiguous values (e.g., X in Gender) for ensuring data consistency.Furthermore, we used the duplicated() method, and we confirmed that no duplicate rows existed in the dataset, which maintained its uniqueness. To evaluate relationships among features, we employed correlation heatmap, which highlighted the predictive power of encoded missingness variables, such as IME_4_Missing, First_Hearing_Missing, and C_3_Date_Missing, in relation to the target variable. Finally, to streamline the dataset, we removed redundant columns where missingness had been encoded.

2.1.2. Inspecting and Transforming Categorical Values

In this part, we focused on data types and encoding categorical variables. First, we converted columns that are object-type into datetime formats for simplifying accurate date-specific computations. Then, we transformed from float to object types to align with their categorical nature, which were numeric columns representing categorical codes. We also converted continuous numeric features from float to integer, reflecting their discrete values and increasing computing efficiency. Specifically, we encoded Y/N as 1/0 and transformed the 'Gender' column into binary format (M as 1, F as 0) and then we dropped the original 'Gender' column to streamline the dataset. We applied these transformations for both training and test datasets, ensuring consistency across all data subsets. This step was important to prepare the data for further steps.

2.1.3. Detecting Outliers

For handling outliers effectively, we used two methods: the IQR Rule and Winsorization, as well as the Manual Thresholding technique.

At first, we applied the IQR rule to systematically detect and exclude outliers by defining bounds based on the 25th (Q1) and 75th (Q3) percentiles. And we can notice this approach was problematic since it removed (83.89% of rows) too many observations.

To overcome the limitations of the IQR method, we then used manual thresholds based on domain knowledge and visual observations. Specifically, Age at Injury values above 100 years and Average Weekly Wage values above $8,000 were capped to ensure realistic data representation. This approach successfully removed only 0.02% of rows, preserving the dataset’s integrity.

Then, we used Winsorization technique and it replaced the nearest acceptable thresholds instead of removing them. For instance, any Age at Injury exceeding 100 was replaced by 100, and weekly wages over $8,000 were replaced by $8,000. Which proved to be a robust and efficient solution compared to the IQR method and manual filtering, as it preserved the dataset's integrity and ensured better generalisability for analysis and modelling.

2.2. Data Visualization

We employed a variety of summary statistics and visualizations to analyze the distribution of variables and uncover potential trends and correlations. These efforts provided deeper insights into the patterns within the dataset and highlighted critical aspects for further analysis.

- **Distribution of Numeric Features**: Most numeric columns, such as "Age at Injury" and "Average Weekly Wage," displayed distributions with means and medians closely aligned, suggesting near-normal distributions. However, some features exhibited skewness and variability that required additional preprocessing.
- **Univariate Outliers**: Boxplots and statistical summaries revealed potential outliers in features such as "Age

at Injury" and "Average Weekly Wage." For instance, "Average Weekly Wage" contained extreme values well beyond the interquartile range, which could affect the analysis if not addressed.

- **Bivariate Relationships**: Using correlation matrices, we identified notable relationships among features. For example, there was a moderate positive correlation between "Average Weekly Wage" and "Claim Identifier." Such relationships provided insights into the interaction between variables, guiding feature engineering.

These visualizations and statistical summaries formed a critical step in understanding the dataset's underlying structure and preparing it for subsequent modeling and analysis.

3. Data Preprocessing

In this stage, the dataset was prepared for analysis and modelling through a series of data cleaning and transformation steps. Our primary goal was to ensure the data was consistent, free of errors, and ready for further analysis. We identified and resolved inconsistencies in date columns such as Accident Date, C-2 Date, C-3 Date, Assembly Date, and First Hearing Date. Missing values were strategically imputed using related features, and irrelevant or inconsistent columns like Birth Year and Number of Dependents were removed. This preprocessing process also included handling class imbalances, encoding categorical variables, and normalising numeric fields. The data was also split into training and test sets using the train_test_split function. Standardisation was also applied for metric features. These steps were important since they ensured our data was clean, consistent, and ready for robust and meaningful analysis.

4. Feature Engineering

To enrich the dataset and provide meaningful attributes for modeling, we applied several feature engineering techniques. At first, we worked with date-related features; here we extracted from the Accident Date, including Injury Year, Injury Quarter, and Injury Month. These features capture temporal patterns, such as seasonality and annual trends, which help account for time-based variations in the data. Additionally, we created flags for missing dates for columns such as IME4 Count, C-2 Date, C-3 Date, and First Hearing Date. These binary flags proved to be relevant due to the high correlation of missingness with our target variable. We then created another key feature, Days Between Accident and Assembly, which was derived by calculating the time lapse between the Accident Date and the Assembly Date. This feature captures the sequence and duration of events, which may influence the outcome of claims. Furthermore, for the Age at Injury column, we created segmentation into predefined categories: Young Adult, Adult, Mid-aged Adult, Older Adult, and Senior. These age groups enable a more granular analysis of demographic patterns and their impact on target variables.

Lastly, regarding Average Weekly Wage feature, we categorised those into income levels: Very Low, Low, Medium, High, and Very High. As a result, this analysis makes it easier to interpret the influence of income levels on predictions. Collectively, these engineered features enhance the dataset by introducing new dimensions that improve the model's ability to identify patterns and relationships within the data

5. Modeling

The modeling process was designed with the intent of balancing model complexity, interpretability, and predictive accuracy, ensuring the selected model generalizes well to unseen data while addressing the nuances of the dataset.

To explore a range of modeling approaches, we selected three candidates, each chosen for specific reasons:

1. **Logistic Regression**: This was chosen as a baseline model due to its simplicity and interpretability. It provides a good starting point to understand whether the dataset exhibits linear separability.
2. **Random Forest**: A tree-based ensemble model known for its ability to handle non-linear relationships and interactions between features. Random Forest was included to explore its potential for capturing more complex patterns.
3. **XGBoost**: A gradient boosting model that iteratively minimizes error. XGBoost was chosen for its proven effectiveness in handling multiclass classification tasks, particularly when dealing with imbalanced datasets and non-linear relationships.

5.2 Modeling Strategy

5.1.1 Data Preparation

The dataset underwent several preprocessing steps to ensure compatibility with the selected models and enhance model performance:

- **Feature Selection**: Features were selected based on their relevance to the target variable. This was informed by exploratory data analysis, domain knowledge, and statistical correlation.
- **Scaling**: Numeric features were scaled using MinMaxScaler. This step was particularly important for models like Logistic Regression and XGBoost to ensure all features were on a comparable scale, avoiding bias toward features with larger magnitudes.
- **Handling Missing Values**: Missing values were addressed through imputation strategies. For example, mode imputation was used for categorical variables, and mean imputation was applied to numeric features, ensuring no loss of data due to missing entries.

5.1.2 Model Assessment

To ensure reliable evaluation, we employed **Repeated K-Fold Cross-Validation** with 6 splits and 2 repeats. This strategy mitigates the risk of overfitting and provides a robust estimate of

model performance. The **F1 Score (macro average)** was selected as the primary evaluation metric because it balances precision and recall across multiple classes, making it particularly suitable for multiclass classification tasks.

5.3 Results and Observations

The performance of each model was evaluated, and the following results were observed:

1. **Logistic Regression**:
    - **Train F1 Score**: 0.36
    - **Test F1 Score**: 0.36
    - **Analysis**: The low F1 scores indicate that Logistic Regression struggled to capture the relationships in the data. This result was expected given the model's linear assumptions and the complexity of the dataset. Logistic Regression's inability to handle non-linear relationships made it unsuitable for this task.
2. **Random Forest**:
    - **Train F1 Score**: 1.00
    - **Test F1 Score**: 0.75
    - **Analysis**: The model achieved a perfect F1 score on the training data, signaling overfitting. While the test F1 score of 0.75 is promising, the gap between train and test performance raises concerns about its reliability on unseen data. This suggests that the Random Forest model memorized the training data instead of generalizing to the underlying patterns.
3. **XGBoost**:
    - **Train F1 Score**: 0.76
    - **Test F1 Score**: 0.75
    - **Analysis**: XGBoost demonstrated balanced performance with closely aligned train and test F1 scores. This indicates that the model generalized well to unseen data without overfitting. Its iterative learning process and ability to handle imbalanced datasets likely contributed to its robust performance.

5.4 Selected Model

After evaluating the results, **XGBoost** was selected as the final model for deployment. Its ability to balance train and test performance while handling the complexities of the dataset made it the most reliable choice. Unlike Random Forest, XGBoost avoided overfitting, and its superior generalization capability positioned it as the optimal model for this task.

6. Performance Assessment

Performance assessment focused on evaluating the ability of the models to address the problem of predicting Claim Injury Type across multiple classes. The evaluation was conducted using metrics tailored to the nature of the task and the dataset.

6.1 Evaluation Metrics

The following metrics were utilized to assess the models:

- **F1 Score (Macro Average)**: This metric provided a balanced evaluation of precision and recall across all classes, making it ideal for multiclass classification.
- **Confusion Matrix**: Used to visualize the distribution of predictions and identify specific classes where the model struggled or excelled.

6.2 Detailed Observations

1. **Logistic Regression**:
    - Despite its simplicity, Logistic Regression was unable to capture meaningful patterns in the data, as evidenced by its uniformly low F1 scores of 0.36 for both train and test. This result underscores the limitations of linear models in complex classification tasks, particularly when dealing with non-linear relationships and imbalanced datasets.
2. **Random Forest**:
    - The model's perfect train F1 score (1.00) indicates severe overfitting. While its test F1 score of 0.75 is respectable, the disparity between train and test performance raises concerns about its robustness. Random Forest effectively captured patterns in the training data but failed to generalize well to new data.
3. **XGBoost**:
    - XGBoost achieved an F1 score of 0.76 on the training data and 0.75 on the test data, demonstrating balanced and reliable performance. The small difference between train and test scores indicates that the model generalized well. Its handling of imbalanced data and iterative error minimization contributed to its success.

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6.5 Strengths and Weaknesses of XGBoost

Strengths:

- **Generalization**: The alignment of train and test F1 scores highlights the model's ability to generalize effectively.
- **Flexibility**: XGBoost's capacity to capture non-linear relationships and handle imbalanced data made it a strong candidate for this task.
- **Feature Importance**: The model provided valuable insights into feature importance, enabling further analysis of key predictors.

Weaknesses:

- **Complexity**: XGBoost requires careful parameter tuning and is computationally more expensive than simpler models like Logistic Regression.
- **Interpretability**: The model's complexity makes it less interpretable, which could limit its use in regulatory or high-stakes environments.

7. References

[1] Alex Galea (2020). “The Applied Data Science Workshop.” Packt.Get started with the applications of data science and techniques to explore and assess data effectively.

[2] Geron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*.This book provides practical, hands-on guidance on using popular machine learning libraries in Python, including scikit-learn, Keras, and TensorFlow.

[3] Alpaydin, E. (2010). “Introduction to Machine Learning”. This book provides a comprehensive introduction to machine learning, including the basics of supervised and unsupervised learning, decision trees, and ensemble methods.

[4] OpenAI (2024). "ChatGPT: A Conversational AI System." Accessed December 22, 2024, from https://openai.com.

8. Appendix

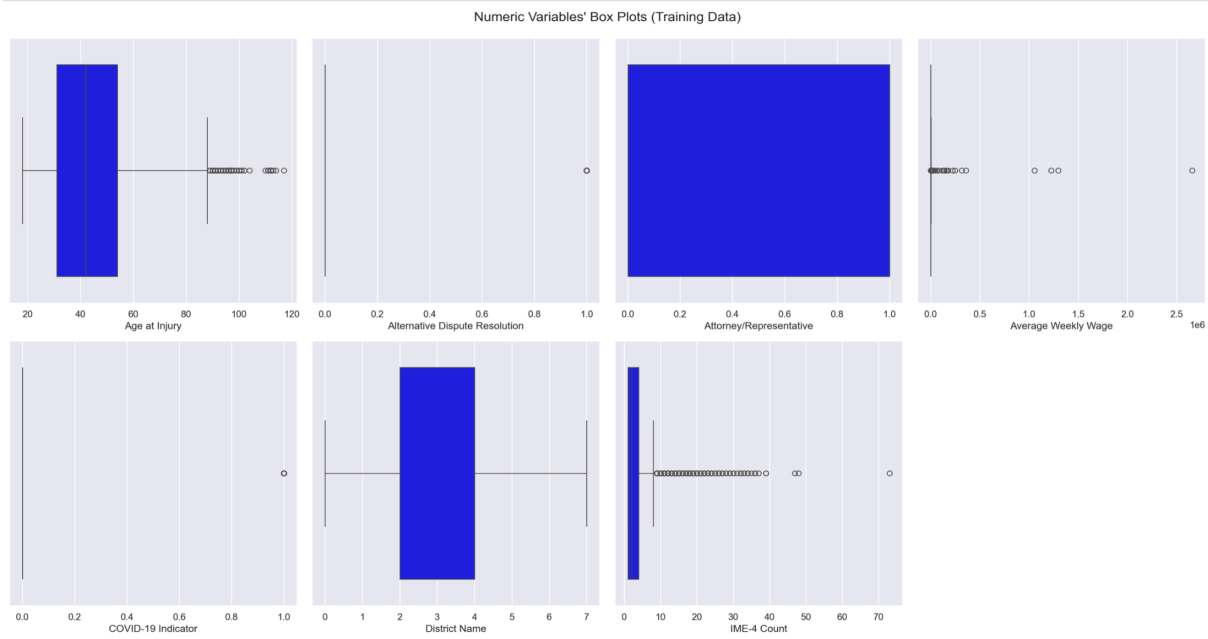


Figure 01 - Numeric Variabes' Box Plots

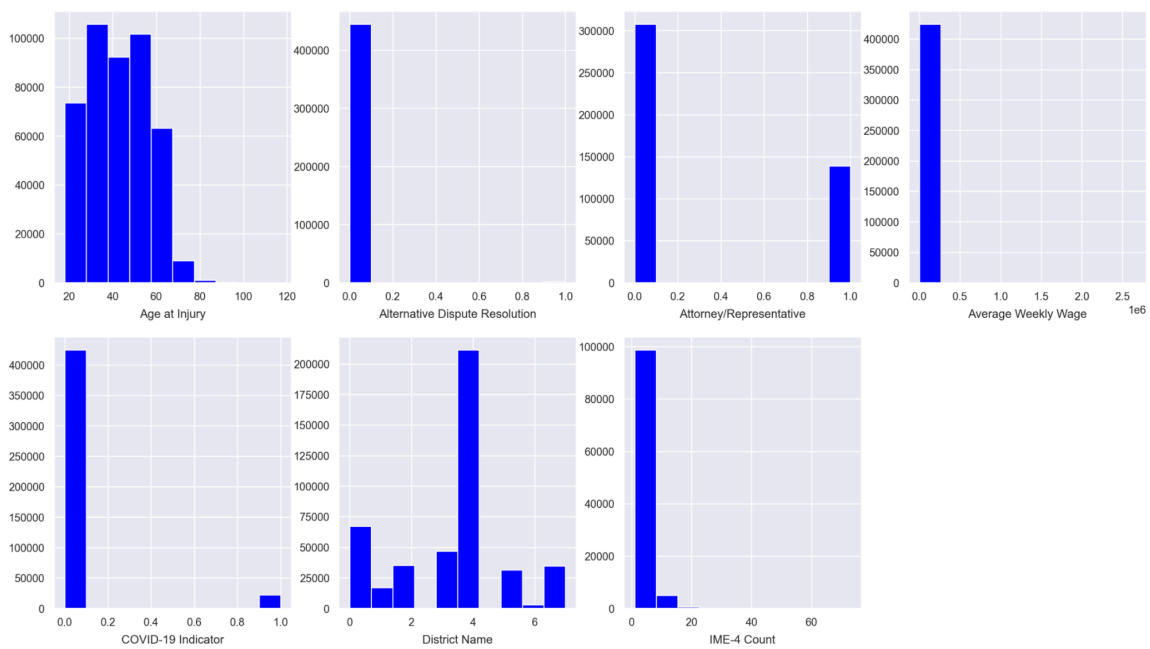
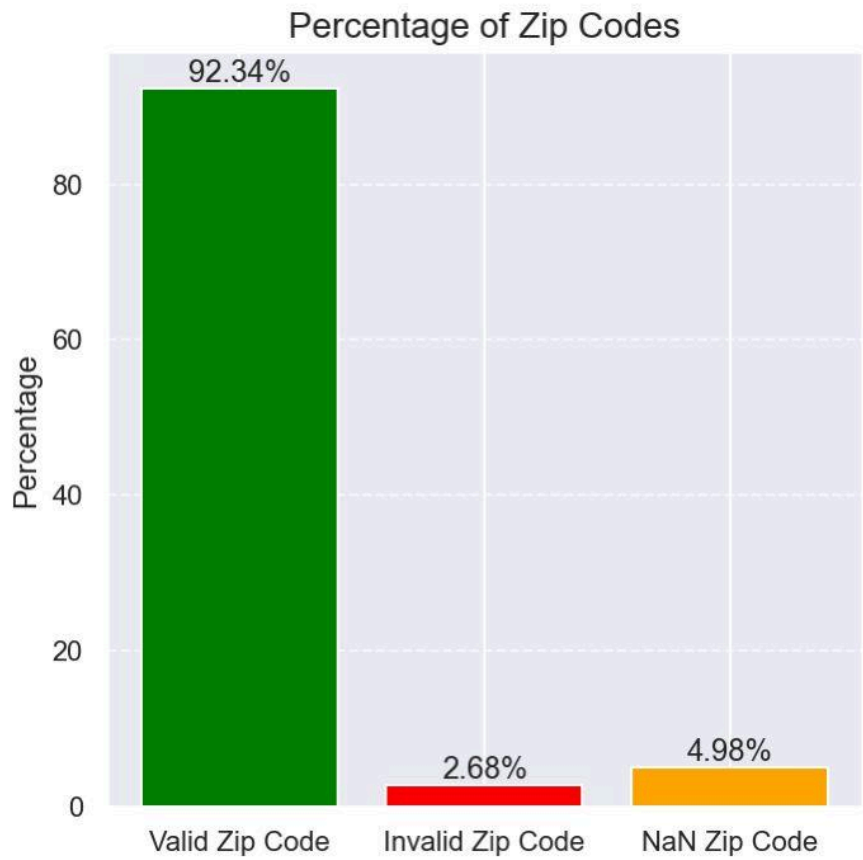


Figure 02 - Feature Distributions



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	Category	Count	Percentage
0	Valid Zip Code	515688	92.337278
1	Invalid Zip Code	14955	2.677790
2	NaN Zip Code	27840	4.984932

Figure 03 - Zip code distrivutions

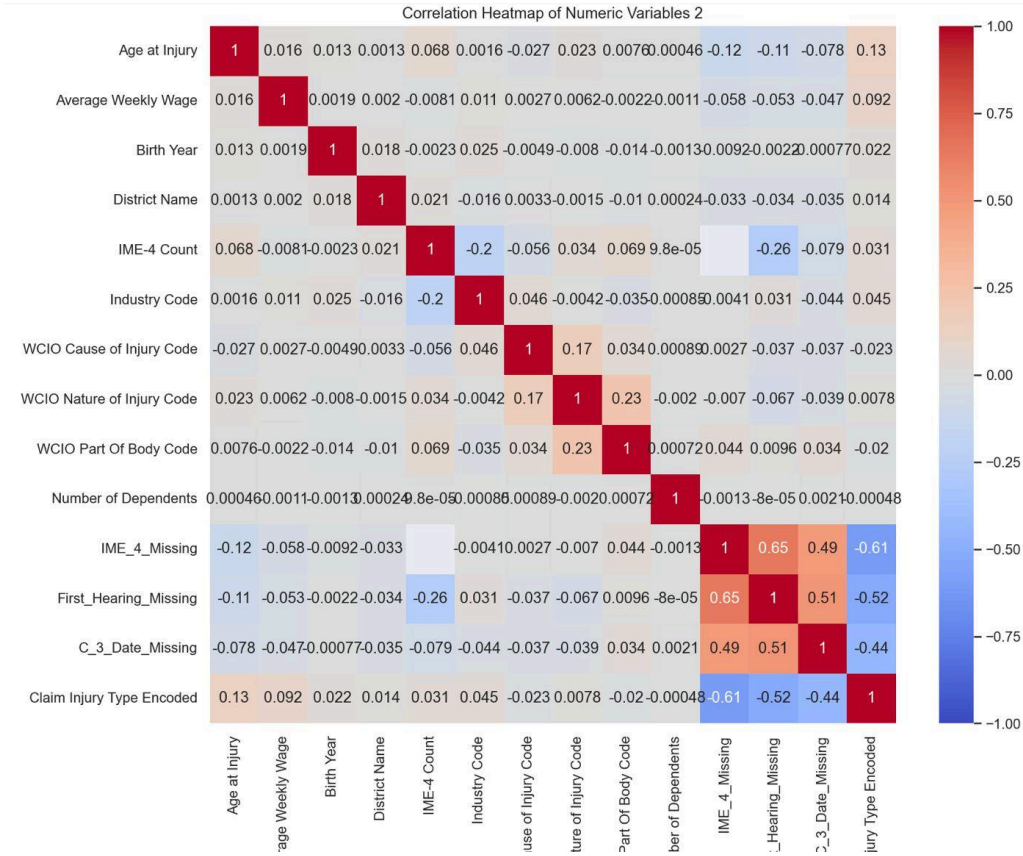

Figure 04 - Correlation Heatmap of variables

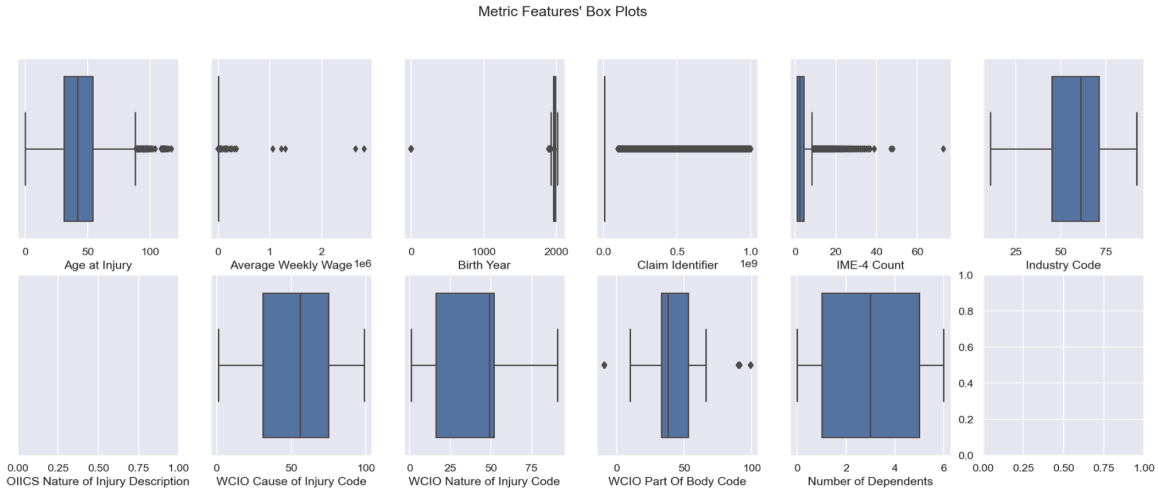

Figure 05 - Boxplots of the variables

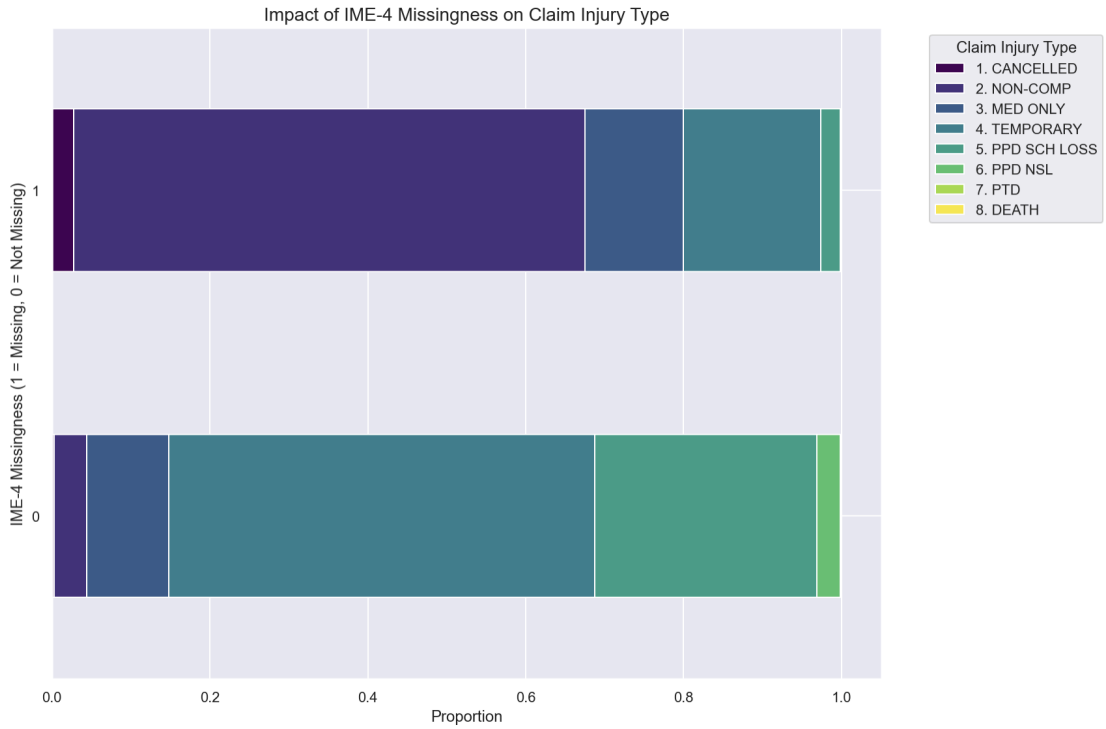

Figure 06 - Impact of IME-4 Missingness on Claim Injury Type

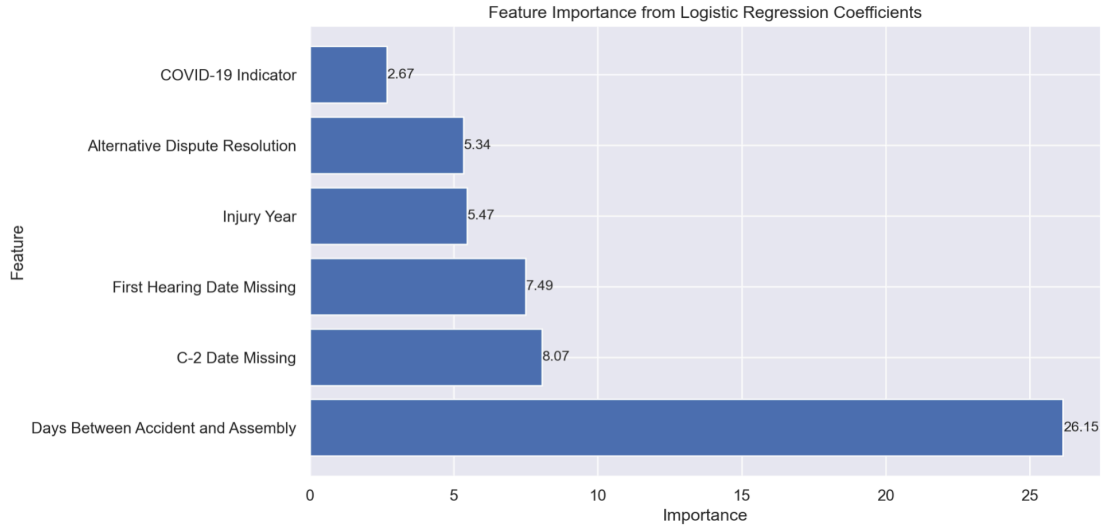

Figure 07 - Feature importance from Logistic Regression Co-efficients

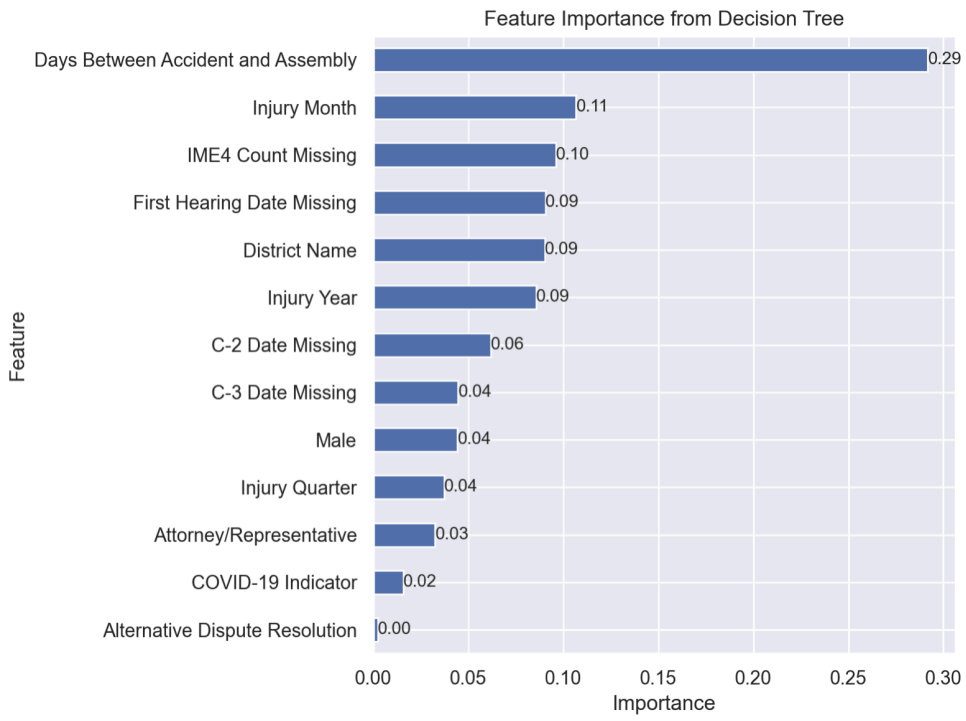

Figure 08 - Feature Importance from Decision Tree